



## **Bogdan Plaiasu**

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*Senior Data Engineer with 12+ years of experience in designing enterprise data platforms, building scalable ETL pipelines, and modernising data architectures. Databricks Certified Data Engineer Professional specialising in large-scale data platforms, Spark optimisation, ETL architecture, and cloud data engineering. Proven track record leading migrations, improving pipeline performance, and building reusable engineering frameworks.*

### **Core Competencies**

Data Engineering • Databricks Lakehouse • Apache Spark • SQL Optimisation • Python • ETL Architecture • Airflow • Azure • Snowflake • Data Quality • Data Warehousing • Performance Tuning • Technical Leadership

### **Key Achievements**

- Databricks Certified Data Engineer Professional
- Designed reusable testing framework adopted across all squads
- Led migration of legacy pipelines to Databricks
- Built enterprise data validation framework
- Technical lead for enterprise SSOT initiatives

### **Tech Stack**

- Languages: Python, SQL
- Data Platforms: Databricks, Spark, Azure
- Orchestration: Airflow, ADF
- Frameworks: PyTest, Django, Laravel
- Databases: MSSQL Server, MySQL, Exasol, Snowflake

# Professional Experience

## SENIOR BIG DATA ENGINEER | COGNIZANT

02/2021 – 08/2026

### DPD (days past due) module optimisation

Led performance optimisation and testing improvements for the DPD module, a critical component used in financial risk and credit analysis.

- Reduced execution time of critical DPD pipelines by 45%, enabling faster financial reporting and lowering compute costs.
- Analysed bottlenecks across data pipelines and implemented optimisations in Spark/SQL processing and data transformations
- Enhanced scalability and reliability of the module to support large data volumes and financial reporting needs
- Contributed to unit and functional testing strategy, improving validation coverage and reducing defects

Stack: Databricks • PySpark • SQL

### Enterprise Data Validation Framework (Non-Deterministic Checks)

Designed and built an end-to-end data validation module to detect non-deterministic changes between pipeline runs, enabling consistent data quality verification.

- Defined comparison strategy to detect unintended differences between pipeline executions
- Implemented ingestion, alignment and row/column-level difference logic
- Built rule-based filtering to isolate non-deterministic discrepancies
- Delivered structured outputs for downstream validation tools and dashboards

*Business Impact:*

- Reduced false-positive data discrepancies by 100% through rule-based filtering
- Enabled automated validation for pipelines

Stack: Python • Databricks • SQL • PySpark

### Enterprise Python Data Testing Framework

Designed and implemented a reusable, modular Python testing framework adopted across multiple teams to standardise data quality practices.

- Architected a class-based testing framework with modular components
- Integrated PyTest for unit and functional testing
- Packaged solution using pyproject.toml with versioning and dependency management
- Provided documentation and onboarding materials to accelerate adoption

#### *Business Impact:*

- Adopted by 8 engineering squads within six months, standardising data quality practices across the organization
- Standardised testing practices across teams

*Stack:* Python • PyTest • OOP • Packaging (PEP 621)

### **Enterprise Retail Data Platform**

Designed and implemented a centralised data platform (Single Source of Truth) supporting analytics and reporting.

- Designed and implemented production-grade Airflow orchestration supporting enterprise data pipelines
- Designed scalable transformation logic powering enterprise reporting
- Led migration of legacy pipelines to the Databricks Lakehouse platform.

*Stack:* Airflow • Snowflake • PySpark

### **SSOT (Single Source of Truth) for a food industry client**

Designed and implemented a unified data architecture integrating multiple heterogeneous data sources.

*Stack:* Azure Data Factory (ADF) • Airflow • Snowflake • Azure DevOps

## **DATA ENGINEER | AERA TECHNOLOGY**

07/2019 – 02/2021

### **Data Warehouse Implementation**

Developed ETL pipelines and optimised SQL workloads for high-performance data processing.

- Developed Python-based ETL pipelines
- Optimised complex SQL queries for performance
- Collaborated with stakeholders to refine requirements

*Stack:* Exasol • Python • DWH 2.0 (Data Vault)

## **SENIOR DATA ENGINEER | SMART-X NET APPS**

07/2017 – 06/2019

### **Database Architecture for Fraud Detection application**

Designed a data warehouse and logic for fraud detection using decision-tree-based rules.

- Designed and implemented database architecture (snowflake schema)
- Built pipelines and logic for fraud detection
- Integrated BI tools for reporting

*Stack:* MySQL • Python • Docker • SiSense (BI tool)

## **SENIOR PROGRAMMER & Team Leader | EMAG**

04/2015 – 07/2017

### **Retail Monitoring Data Warehouse**

Designed and implemented a data warehouse supporting business reporting and operational monitoring.

- Modelled database using snowflake schema
- Built data ingestion pipelines across multiple sources
- Developed dashboards for business users

Stack: Database design (DDL, normalisation) • MSSQL Server - Queries • Optimisations and Stored Procedures • PHP • Ajax

## **Earlier Career (Software Engineering & Web Development)**

Strong foundation in application development

- ERP systems development and maintenance
- Web application modernisation
- SQL development and reporting solutions

## **Certifications**

- Databricks Certified Data Engineer Professional | Databricks | 2026
- Databricks Certified Data Engineer Associate | Databricks | 2026
- Certified Azure Fundamentals | Microsoft | 2024
- Developer Foundation | Databricks | 2021
- Hands On Essentials | Data Warehouse by Snowflake | 2021
- Exasol Essentials | Exasol | 2020
- Optimal Motivation | Human Invest | 2016
- Situational Leadership | Human Invest | 2016
- Essentials for Managers | Human Invest | 2016
- PHP Certification | InfoAcademy | 2011

## **Education**

- Bachelor of science, Aerospace - Degree in "Propulsion Systems and Rockets" | Polytechnics University of Bucharest | 2003